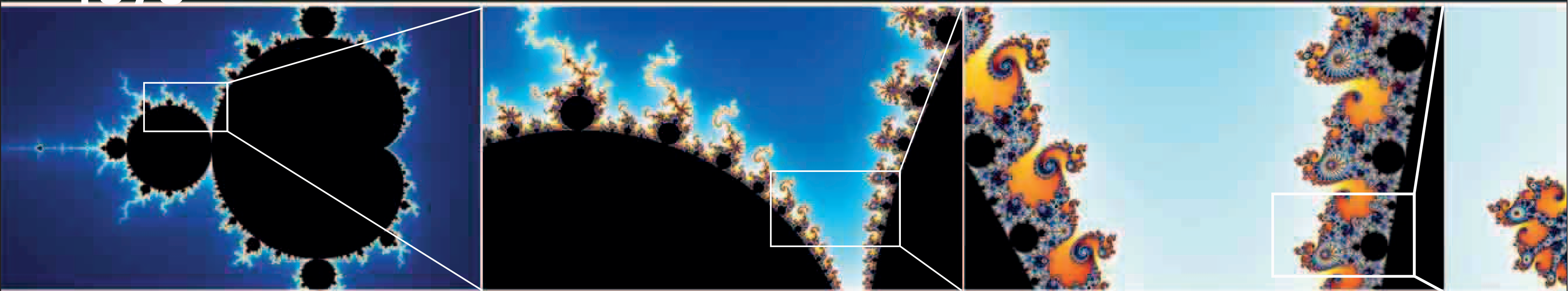

The Mandelbrot set is a simple mathematical formula that, when visualized using computer graphics, reveals intricate beauty at different scales.

IN JUNE 1975, Japanese electronics company Sony released the first home video recorder format, **Betamax**. **Video cassette recorders** (VCRs) allowed people to record television programs and watch prerecorded films rented or bought from video stores. Although VCRs had been available since the 1960s, they were expensive and very few households owned one. Betamax was inexpensive enough to be purchased by many families for home use. The following year, another Japanese company, JVC, came out with a rival format—**VHS** (Video Home System). Over the next decade, the two systems competed in a "format war"; by the end of the 1980s, Betamax VCRs and cassettes represented only about 5 percent of the home video market, although they remained the standard for broadcasters and professional video editors until the rise of digital video production.

In July, millions of people around the world watched a historic moment live on their televisions: the docking of a **Russian Soyuz spacecraft** with an **American Apollo** module. The two craft were together in orbit for more than 40 hours, during which the crews carried out joint experiments, exchanged gifts, and un-docked and re-docked their craft several times.

Farther out in space, the Russian unmanned space probe Venera 9 became the first spacecraft to **orbit Venus**. A spherical entry pod detached from the orbiter and opened to release a lander probe, which descended to the planet's surface. It was the first probe to **send images back from the surface of another planet**; the orbiter acted as a relay station between Venus and astronomers on Earth. The lander relayed data and images for 53 minutes, after which radio contact was lost.

In November, Polish–French mathematician **Benoit Mandelbrot** (1924–2010) coined the word **fractal** in his book *Les Objets Fractals: Forme, Hasard, et Dimension* (*Fractals: Form, Probability, and Dimension*). The mathematics of fractals provided a way of bringing apparently **rough, irregular, and chaotic phenomena** into the domain of mathematics. It allowed mathematicians to understand and model intricate natural forms

53 MINUTES

THE LENGTH OF TIME THE **VENERA 9 LANDER** OPERATED ON THE **SURFACE** OF **VENUS**

Venera 9 lander
This is a model of the Venera 9 lander probe that carried out tests and sent panoramic photographs from Venus.

COLLABORATION IN SPACE

Since the 1950s, the US and USSR had been engaged in a "space race," each superpower trying to assert its technological superiority. The Russians were the first to put a satellite into orbit, launch a person into space, and send probes to the Moon. Not far behind, the Americans achieved perhaps the biggest coup, by landing astronauts on the Moon's surface. The docking of the two nations' craft in orbit—the Apollo–Soyuz Test Project—was an important statement of peace and collaboration between the US and USSR, a reflection of the easing of tensions between them.

April IBM announces release of the IBM 3800, the first commercially available **laser printer**

April 19 Launch of India's first satellite, **Aryabhata**

June 7 Sony releases the **Betamax** video recording format

July 17 Russian and **American space capsules** dock in orbit

August 8 American geochemist Wallace Broecker coins the term **global warming**

October 22 Venera 9 lander **touches down** on the surface of **Venus**

November 18 Benoit Mandelbrot coins the term **fractal**